

ROBOTICS



We will talk about;

- **What is a robot?**
- **Types of Robots**
- **Uses of Robots**
- **Challenges/disadvantages of using Robots**
- **Basic components of Robots**
- **Steps to construct robot from tool kit**





WHAT IS A ROBOT?

- It is a machine programmed by a computer capable of carrying out different task.



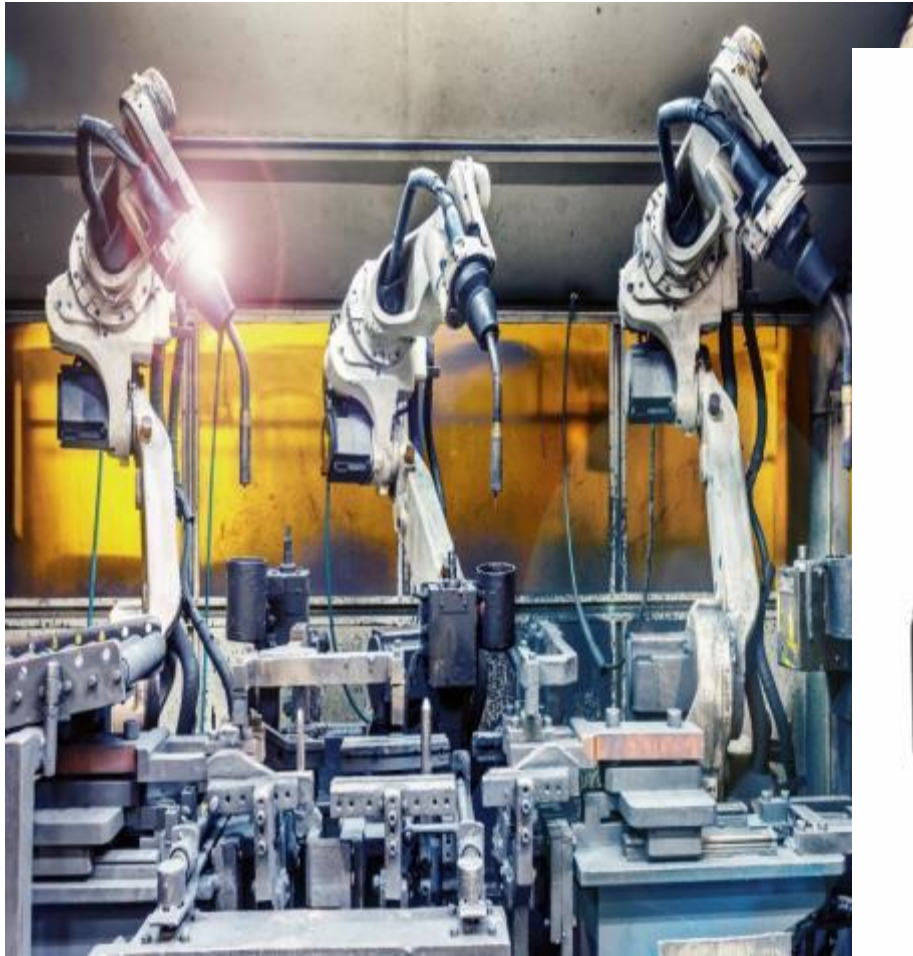
TYPES OF ROBOTS

- **FIX ROBOTS**

They are robots positioned a particular place to perform a specific function or to solve a problem. They are mostly called *industrial robots* or *stationary robots*.

Some examples are, gantry robots, parallel robots, robotic arms etc

SOME PICTURES OF FIXED ROBOTS



MOBILE ROBOTS

They are movable robots that can move around their environment using wheels or legs.

Some examples are *humanoid robots, hospitality robots*, etc



PICTURES OF MOBILE ROBOTS





APPLICATIONS OF ROBOTIC MACHINES IN SOCIETY

- In education

It develops learners skills by developing their knowledge through the creation, design, assembly and operation of robots.

It is used now in the classroom for learning and teaching purposes and extra curriculum.



APPLICATION OF ROBOTICS

- **IN AGRICULTURE**

In farming, robots can be used for hand tilling, harvesting and weeding.

Robots can be used for thinning out of seedlings on the farm.

Robots are used to monitor the health of farm animals.

- **IN MILITARY**

They are used for scanning and searching the environment for incoming attacks

They use drones (robots) to monitor the environment

They are used to carry military supplies.

APLLICATIONS OF ROBOTS/ USES

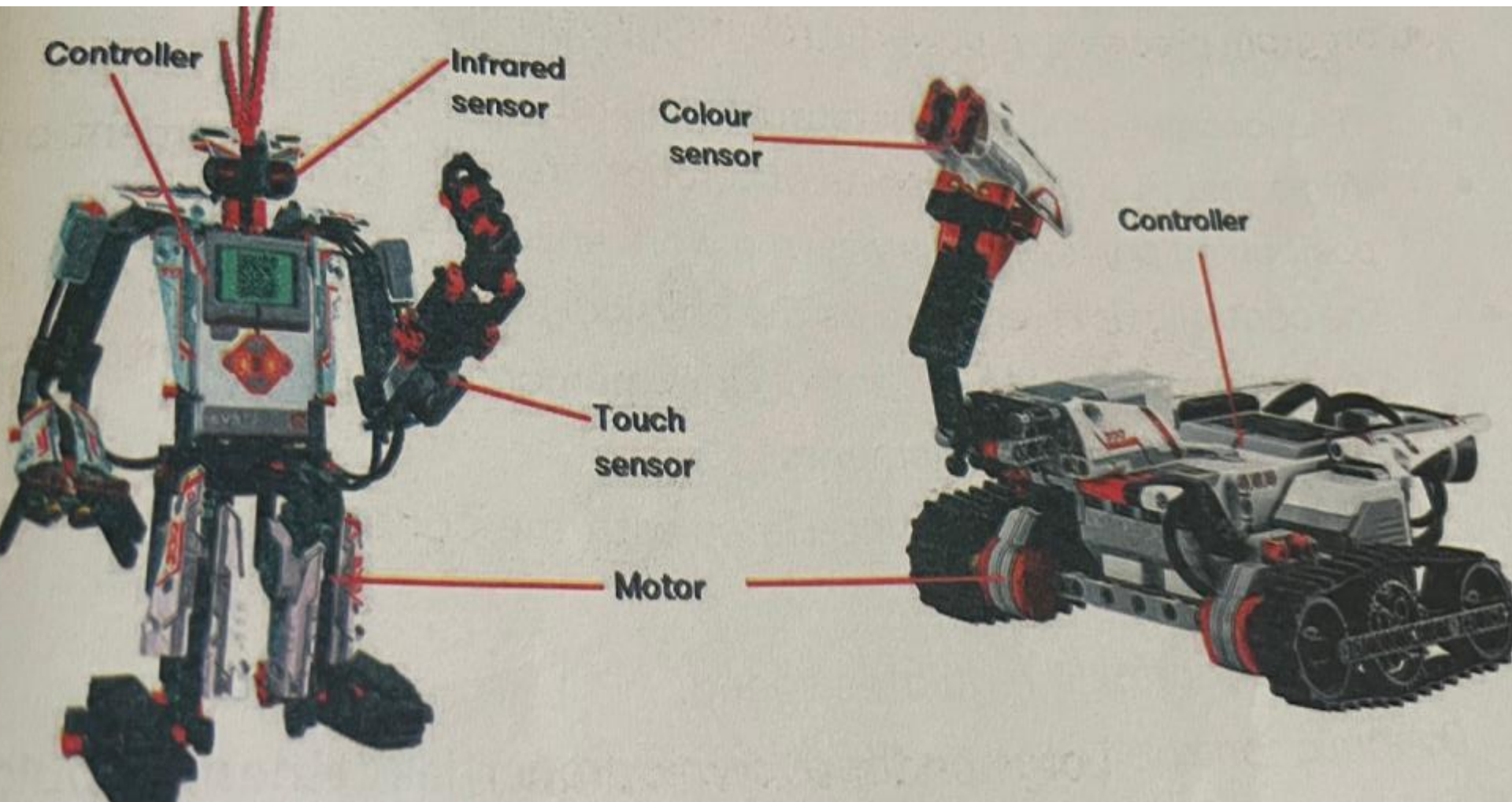




3 BASIC COMPONENTS OF ROBOTS

- **Controller**
- **Mechanical**
- **sensor**

PARTS OF AN EV3 ROBOTS



CONTROLLER OF A ROBOT



It is the brain of the robots which receives data from sensors, processes it using programmed instructions and sends commands to the actuators to make the robot act. It is usually a microprocessor or a micro controller like arduino.

Types of controllers based on their function

Kinematic controller- it controls a robots position, velocity and path.

Dynamic controllers- it controls a robots by considering the force, mass and friction which affects motion.

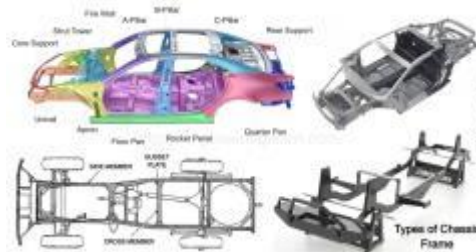


FUNCTIONS OF A CONTROLLER OF A ROBOT

- It reads data from sensors
- It runs logic programs
- It sends signals to actuators
- Coordinates multiple movement

MECHANICAL COMPONENT

- They are the physical parts of the robots that provides structure and produce motion to perform task.
- It handles all physical movement and structure of the robot. It's the robots skeleton, muscles. If the controller is the brain of the robot, the mechanical parts are the bones, joints and muscles.



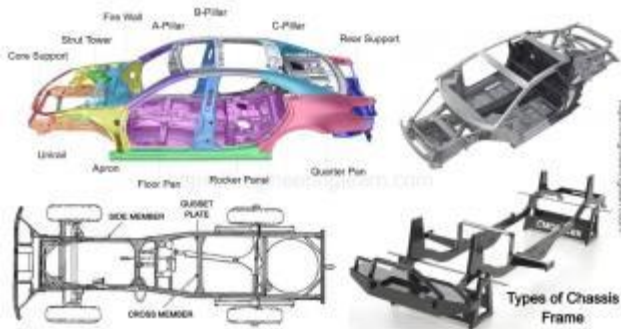


Some mechanical components of a robot are;

- Actuators
- End effectors
- Transmissions
- Locomotion system
- Chassis/ Frame
- Joints

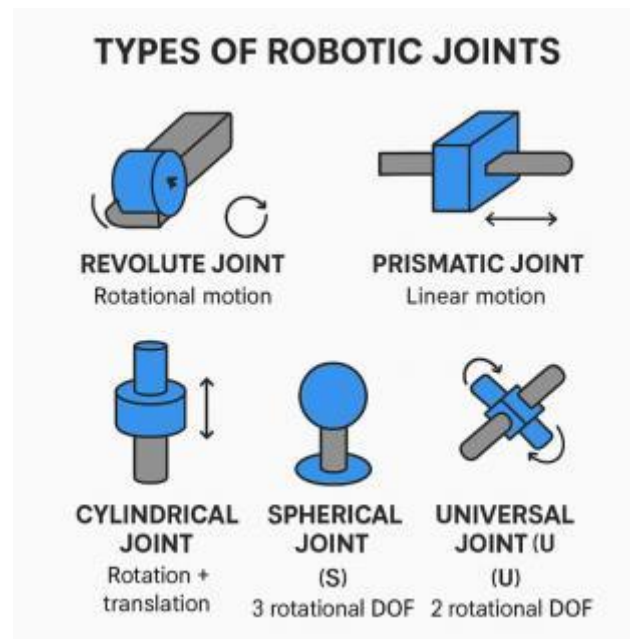
CHASIS OR ROBOTIC FRAMES

- This part of the mechanical component of the robot gives shape and strength to the robot. It must be light but very firm to support the robot.



Joints

- It is connection between parts of a robot that links to allow movement between them.



Actuators OF A ROBOT

- They are devices in a robot that convert energy from the controller into force to move joints, wheels of a robot to enable the robot to perform physical task (changes electrical to mechanical energy)
- **Motors** make the wheels spin and the joints of the robot to rotate.
- **Pistons** helps the robot in a straight motion.



MOTORS



PISTONS

End Effectors OF A ROBOT

- **Grippers** are attached to the end of the robot arms to help it to hold and release objects.



Transmissions OF A ROBOT

- **Gears** are used change the speed of robots.



Locomotion System OF ROBOT

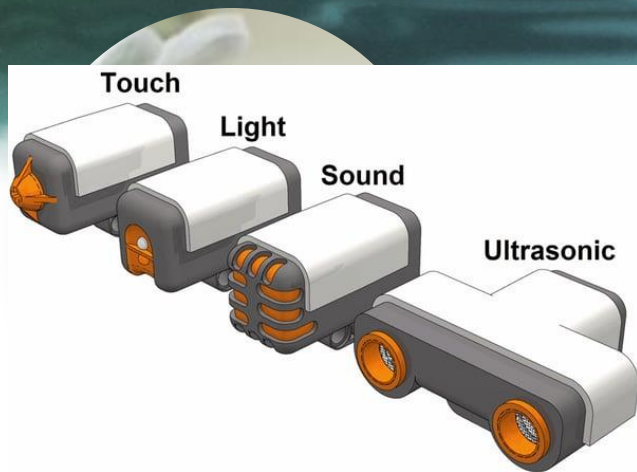
- **Wheels** are used for mobile robot to move.

Robot Wheels:

A Guide for Mobile Robotics

What wheels should you use for your robot?



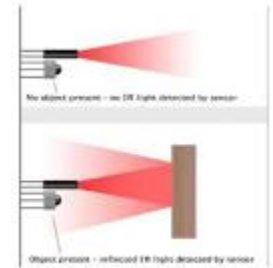
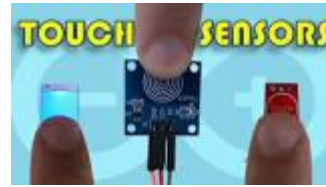


SENSOR OF A ROBOT

It is a device that helps the robot to detect and respond to environmental conditions helping it to interact with the environment.

TYPES OF ROBOT SENSORS

- Touch sensor
- Color sensor
- Infra-red sensor
- Ultra sonic sensor
- Visual sensors
- Audio sensors
- Motion sensors
- Ambient light sensor





APPLICATIONS OF SENORS IN REAL LIFE SITUATIONS

- In Home security, sensors can be used to detect intruders and also to alert owners. Examples of such sensors are motion sensors, cameras, etc.
- In Gaming, motion sensors can used to track and detect players movement.
- Ambient light sensor is used to adjust the brightness of the phone's screen.
- Vehicle detection sensors like inductive loop sensors are used in traffic lights to detects vehicles and pedestrians at the road side.
- Motion sensors can be used to make automatic doors which can open and close by it self.



LAWS OF ROBOTICS

ASIMOV'S THREE LAWS OF ROBOTICS

1. A robot may not injure a human being or, through inaction, allow human being to come harm
2. A robot must obey the orders given to it by human beings, except where such orders will conflict with the first law.
3. A robot must protect its existence as long as such protection doesn't conflict with **First** and **Second** law.



ROBOT SIMULATIONS

It is a kind of software that imitates real-world behavior of robots which allows you to test and develop robots.

BENEFITS OF SIMULATION

1. It is safe because it allows you to test without risking hardware damage.
2. It is less expensive because there is no need for physical hardware of robots.
3. Components of robots could be tested before simulation.



SOME ROBOT SIMULATION SOFTWARES

- Gazebo
- Mbot simulator
- Webots
- Scratch
- Light bot
- EV3 Robot virtual worlds
- Virtual robotics toolkit
- Visual components
- roboDK
- V-REP
- RobotStudio